



**University of
Nottingham**

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Semester 1 Deliverables

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(Group Project Coordinator)**



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State of Play



Module Structure

Semester 1

Project Signup

Competitive Bidding

Project Allocation

Development 1

CV

Proof of Concept

Interim Report

Interim Peer
Assessment

Semester 2

Development 2

Final Product

Demo Day

Repos

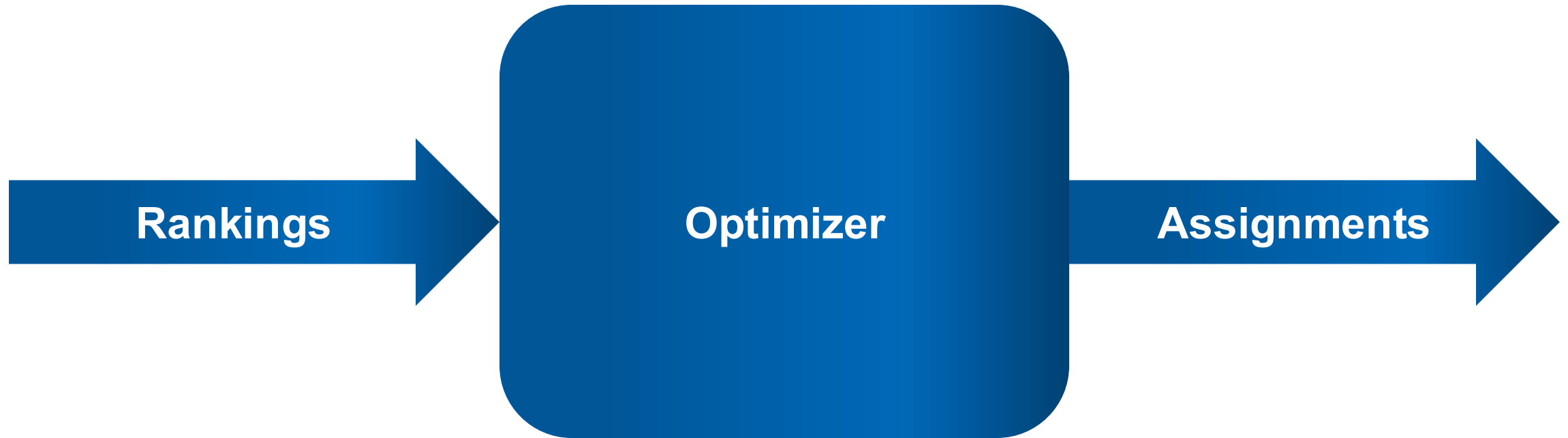
Group Reports

Final Peer
Assessment

Individual
Report



Project Allocation





- **All teams have been allocated to a project**
- **The allocation maximises fairness across all teams**
 - Non-linear penalties
- **Provably optimal solution**
 - Third choices that were allocated were provably unavoidable



- **You should be in contact with your supervisor / sponsor**
 - **Strongly** recommend setting up a recurring meeting with your academic supervisor - no longer than once every 2 weeks.
 - **Every** team member is expected to be present during these meetings
 - **Every** team member should contribute equally during these meetings



Development 1 - Expectations

- **Implement an initial prototype**
 - Demonstrates that your plan works – or learn otherwise!
- **Teams take ownership of their project:**
 - Supervisors provided guidance
 - Expected to use SCRUM process
- **You are expected to use Git**
 - All team members - make sure you have logged on.
 - Git leads – make sure everyone can work conveniently.



- **Admin:**
 - Non-disclosure agreements (NDA)
 - Preliminary ethics checklists
- **Semester 1 deliverables:**
 - Interim report (underpins **development 2** and **final report - rubric**)
 - Demonstration
 - Interim peer review
- **Semester 2 deliverables:**
 - Individual reports
 - Final group reports
 - Code repository
 - Demonstration
 - Final peer reviews



Interim Report

			Criteria	Mark Weight	Awarded Mark (out of 100)
Interim Report	Report & Prototype	Project Background & Understanding (10%)	Criteria: - Evidence of understanding of the initial brief / specification. - Evidence of own interpretations / additions / innovations. - Depth and context of the problem presented (e.g. existing products).	10%	
		Requirements & Critical Analysis (15%)	Criteria: - Depth / clarity / coverage and appropriate prioritisation of requirements. - Motivation / reflection for changes to requirements and scope reductions. - Results of focus groups / user surveys (where applicable). - Analysis of hardware / tools / methods / software libraries for the project.	15%	
		Initial software implementation & testing (30%)	Criteria: - Data storage / management / manipulation / use of (personal) data (if applicable). - Architecture / design / proof of concept of the prototype. - Appropriate use of version control system (branching / merging). - Evidence of testing. - Preliminary inline software documentation.	30%	
		Project Management & progress (15%)	Criteria: - Evidence of effective / appropriate project management (Agile, SCRUM, SCRUMBAN, Kanban, etc.). - Documentation of sprints / retrospectives / workload management / formal meetings (where applicable). - Use of project management tools (Trello / Microsoft Planner / Teams / Slack). - Effective use of team skills / skills transfers / cross training. - Professional attitude / disciplined / consistent / timely / punctual / autonomous & self organising.	15%	
		Preliminary Reflection (15%)	Criteria: - Tools / techniques / methods / technical difficulties / testing. - On project management / current progress. - Functioning of the team so far. - Forward plan / address concerns identified.	15%	
		Style (8%)	Criteria: - Accessibility / clarity & motivation of the argument / structure (e.g. by key parts / components of the project) - Concise / conceptual descriptions / appropriately balanced content / length / depth - Grammar / spelling / articulation / formatting / structure / jargon & definitions - Use of visuals / graphics	8%	
		Resource links (2%)	Criteria: - Links to document repository / project management documentation (e.g. Trello / Microsoft Planner) / code repository	2%	
		Conclusion & Summary (5%)	Criteria: - summary of key points and future plan	5%	
		Total:			



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Mark Scaling Recap



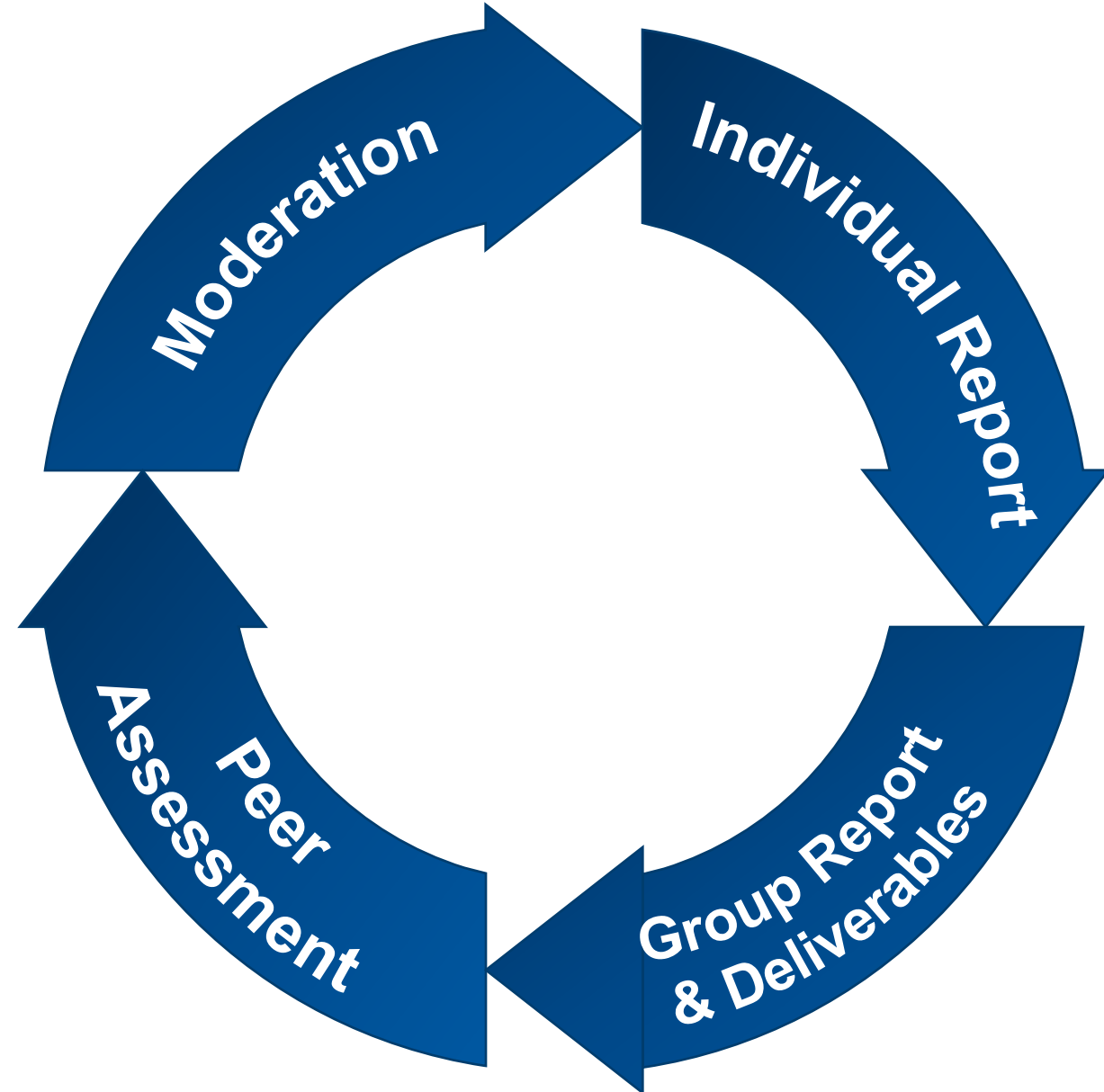
Group Deliverable	Component Contribution	Overall Contribution
Eol & Pitch	15%	12%
Interim Report	10%	8%
Interim Peer Assessment	-	-
Group Report	25%	20%
Stand / Demo / Presentation	15%	12%
Software / Project Management	35%	28%

Individual Deliverable	Component Contribution	Overall Contribution
CV	20%	4%
Peer Assessment	20%	4%
Individual Report	60%	12%



Assessment - Scaling

- **Assessment considers:**
 - Group elements (80%)
 - Individual elements (20%)
- **Group elements are scaled based on:**
 - Individual reports
 - Group deliverables (e.g. document and code repositories)
 - Peer assessment (deviation from average)
 - Supervisor moderation





Assessment – Peer Review

Instructions:

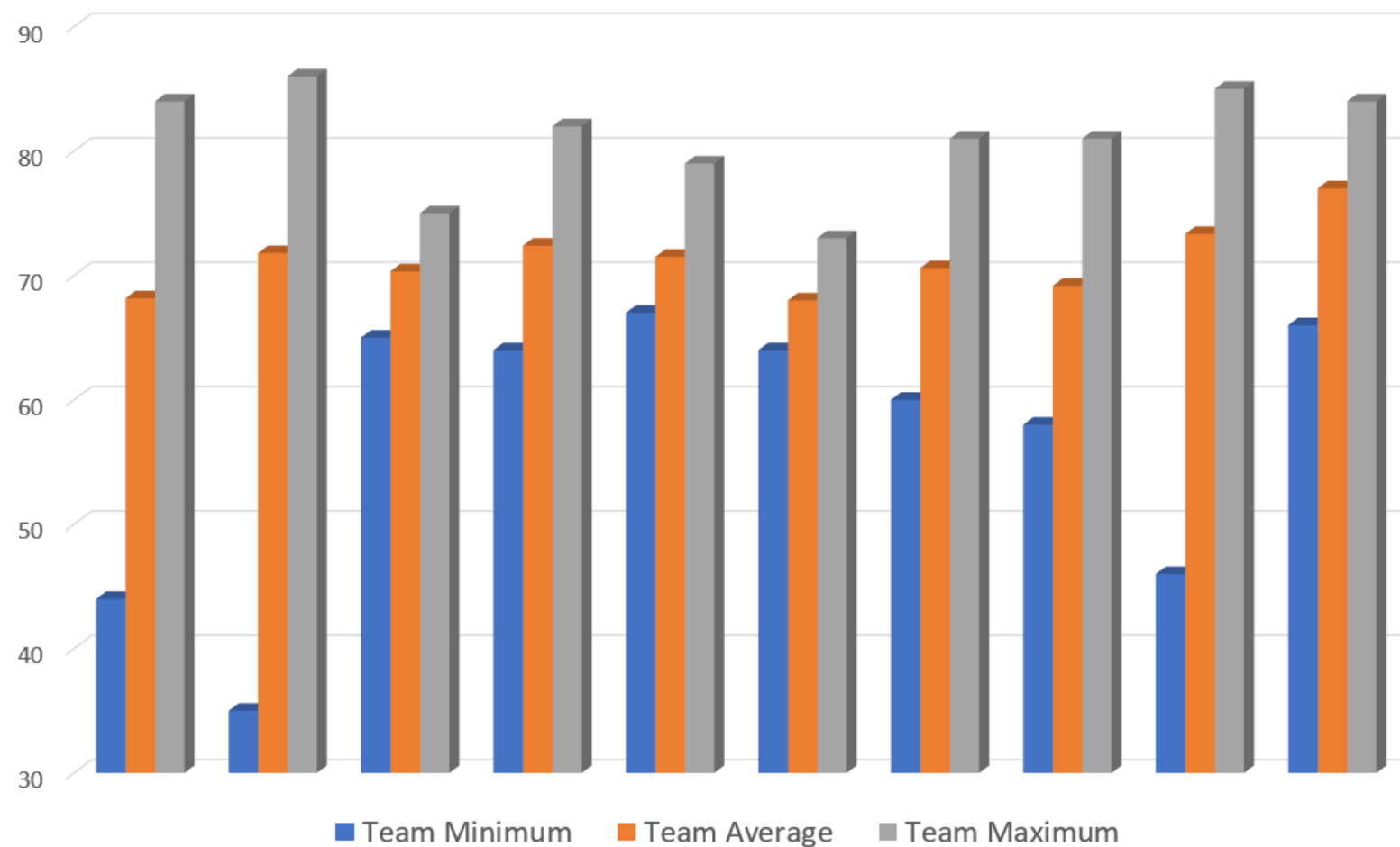
- **Do not change the filename** when submitting it.
- Do not provide peer review scores for yourself.
- Please enter a score between 1 and 5 in the evaluation column for each category. 1 is very poor, 5 is outstanding.
- Please provide a motivation for your score for each of the categories.
- Please leave cells blank for member that have not participated in the module.

<u>Team Number</u>	<u>Last Name</u>	<u>First Name</u>	<u>Username</u>	<u>Category</u>	<u>Evaluation</u>	<u>Motivation</u>
1 Best	Adam	psyab29		General Contribution to the project		
				Attendance at meetings		
				Creativity		
				Work rate		
				How did they perform as a team member		
				Constructive feedback to share		
1 Fisher	Frankie	efyff1		General Contribution to the project		
				Attendance at meetings		
				Creativity		
				Work rate		
				How did they perform as a team member		
				Constructive feedback to share		



Assessment - Scaling

Marks Distributions within Teams





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SCRUM



“To achieve great things, two things are needed: a plan, and not quite enough time.”

Leonard Bernstein



- Plan – Requirements, design, logistics
- Do – Build the bridge
- Show – Release the bridge to the client





How NOT to Build Software



- Plan – Requirements, design
- Do – Coding
- Show – Release software to the client

Surefire route to shocks and nasty surprises



Software Project Challenges

- Extreme levels of complexity
- Stakeholders lack understanding of how software works
- Emerging and evolving requirements
- Uncertainty
- Rapid pace of change
- Less mature discipline



Software Project Opportunities

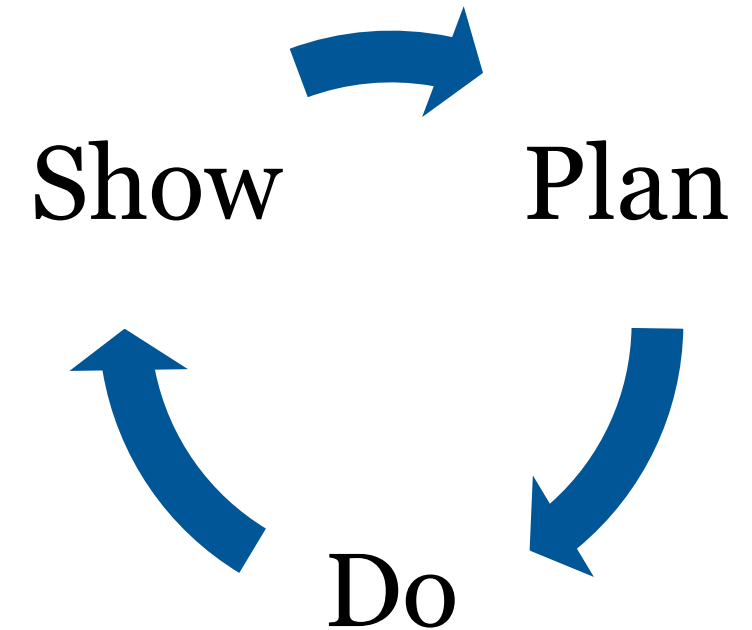
Software is:

- Quick to prototype
- Easy to change
- Modular
- Lots of flexibility
- Development by small teams



A Better Approach

- Iterate a shorter process
- Gather information frequently:
 - Client feedback
 - Decisions
 - Estimates
- Less surprises, better estimation, regular visible progress
- Planning informed by earlier cycles
- Each cycle is a **sprint**





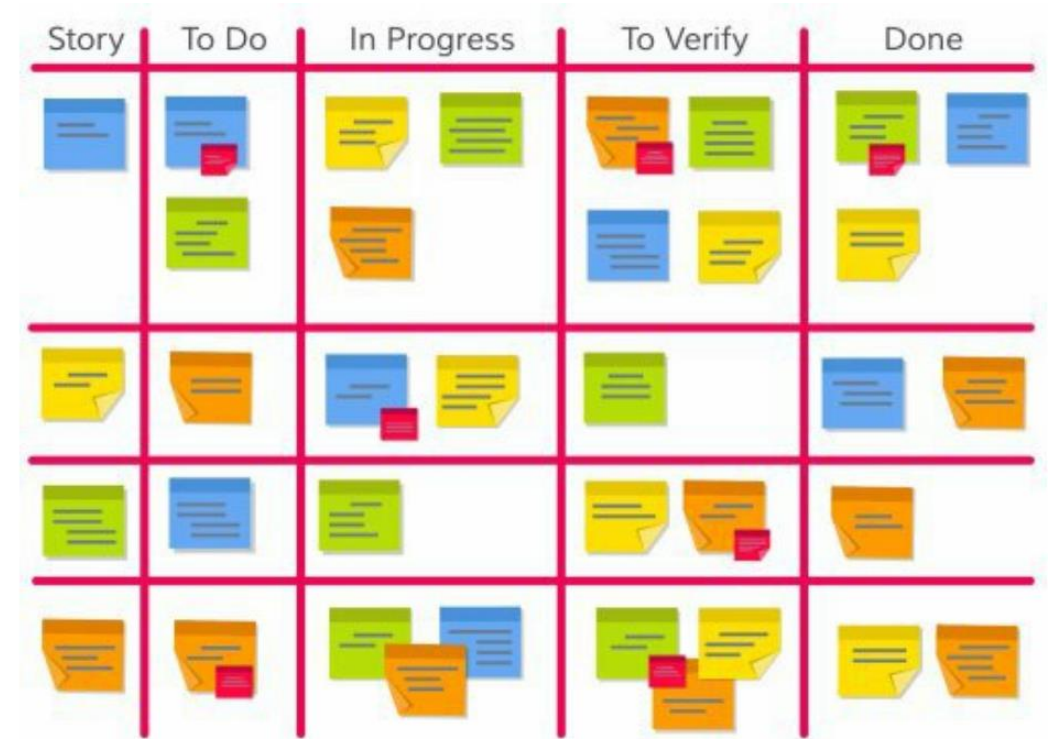
SCRUM – Timings

- **Concepts:**

- Story points: estimation of the effort required to complete a feature
- Velocity: #story points per sprint
- Burndown: Chart of story points vs time

- **Tools:**

- Jira, Azure, Trello, MS Planner
- Generate predictions on completion dates and analyse
- What-if analysis





- **Product backlog:**
 - **User story:** "As a *[role]*, I want *[capability]*, so that I can *[benefit]*"
 - Detail added in conversations
 - Acceptance criteria
 - Size – story points
 - **Tasks:** items that provide indirect value to the project
 - **Bugs:** continuous debugging
- **Sprint backlog:**
 - Contains items from the product backlog to address in the next increment



Three roles:

1. Product owner (sponsor/supervisor) – has the vision
2. SCRUM master (team lead) – manages process
3. Development team



Three artifacts:

1. Product Backlog – user stories, bugs, tasks
2. Sprint backlog – items for next sprint
3. Product



Four timeboxed meetings:

1. Sprint planning – shortlist priority user stories for next sprint
2. “Daily” Standup (SCRUM) – achieved, plan, impediments
3. Sprint review – results & demo with stakeholders (sponsor/supervisor)
4. Retrospective – went well, improvements, changes (team)



Meetings - Sprint Planning

- **Considers:**
 - Define overall sprint goal
 - Capacity planning (member availability, coursework, ...)
 - Select user stories for next increment
 - Breaks down user stories into tasks
 - Estimate the story points per task
- **Should not exceed 1h for 2 weeks sprint**
- **Team decides on items to include, sets short term goals**
- **Results in a total number of story points to include (matching capacity and average velocity)**



Meetings - Stand-ups (SCRUMs)

- **No less than twice per week**
- **For every member:**
 - What have you done since the last scrum?
 - What are you committing to getting done before the scrum?
 - What impediments are holding you back?
- **Time boxed to 15 minutes**
- **Review of the burndown chart**





- **Constructive discussion with stakeholders**
 - What was the goal of the sprint
 - What items were completed
 - What successes and problems
 - Product demonstration
- **Discussion of plan and timeline**



- **Discussion between team members**
 - What went well
 - What could have gone better
 - Potential improvements
 - Ideas to try



Key points

- **Gather feedback regularly**
- Learn from previous iterations
- Keep the process lightweight
- Adapt to your team and circumstances
- Client might have views on the exact process



- <https://scrumguides.org/scrum-guide.html#scrum-definition>
- **Todaro, Dave. The Epic Guide to Agile: More Business Value on a Predictable Schedule with Scrum**
- **Sutherland, Jeff; Sutherland, J.J.. Scrum: The Art of Doing Twice the Work in Half the Time . Random House.**



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Team Work



Team Work – The Aristotle Project (Google)

“Who is on a team matters less than how the team members interact, structure their work, and view their contributions.”

Unimportant

- Friendships
- Mix of introverts and extroverts
- Skill, intelligence or expertise

Important:

- **Equality of conversation**, balance of speaking time
- **Ostentatious listening**, demonstrably paying attention (put the phone down!)

[See: How google built the perfect team](#)

Five key dynamics that set successful teams apart from other teams at Google:

1. **Psychological safety:** Can we take risks on this team without feeling insecure or embarrassed?
2. **Dependability:** Can we count on each other to do high quality work on time?
3. **Structure & clarity:** Are goals, roles, and execution plans on our team clear?
4. **Meaning of work:** Are we working on something that is personally important for each of us?
5. **Impact of work:** Do we fundamentally believe that the work we're doing matters?"



Criteria for psychological safety:

1. Do you struggle to have tough conversations?
2. Do you feel judged and team members disrespect each other?
3. Do you fear asking for or delivering constructive feedback?
4. Are you hesitant about expressing divergent ideas or asking “silly” questions?
5. Do you feel you cannot make mistakes or take risks?
6. Are discussions dominated by a few strong voices marginalizing other perspectives?
7. Are your team members competitive with each other?



- **Bad communication can be passive and active**
 - Communication enables to predict with 91% accuracy whether couples will divorce
- **Four factors in toxic communication:**
 - Criticism
 - Defensiveness
 - Stonewalling
 - Contempt

<i>Characteristics</i>	
<i>Criticism</i>	Attacking, blaming, bullying, dominating, disrespecting, chronic complaining
<i>Defensiveness</i>	Looking for excuses, not taking the other person seriously, deflecting blame, not being open to influence
<i>Stonewalling</i>	Withdrawing from the interaction, turning away, refusing to engage, avoiding, being passive and withholding information
<i>Contempt</i>	Cutting others down, hostile gossiping, undermining, disrespecting, demeaning, mocking, being sarcastic, ridiculing, name-calling, eye-rolling, mimicking



- **Communicate Face-to-face when possible**
 - 34 times more likely to get a positive response when asking something in person
- **Frame the work as a learning rather than an execution problem**
- **Model curiosity by asking a lot of questions**



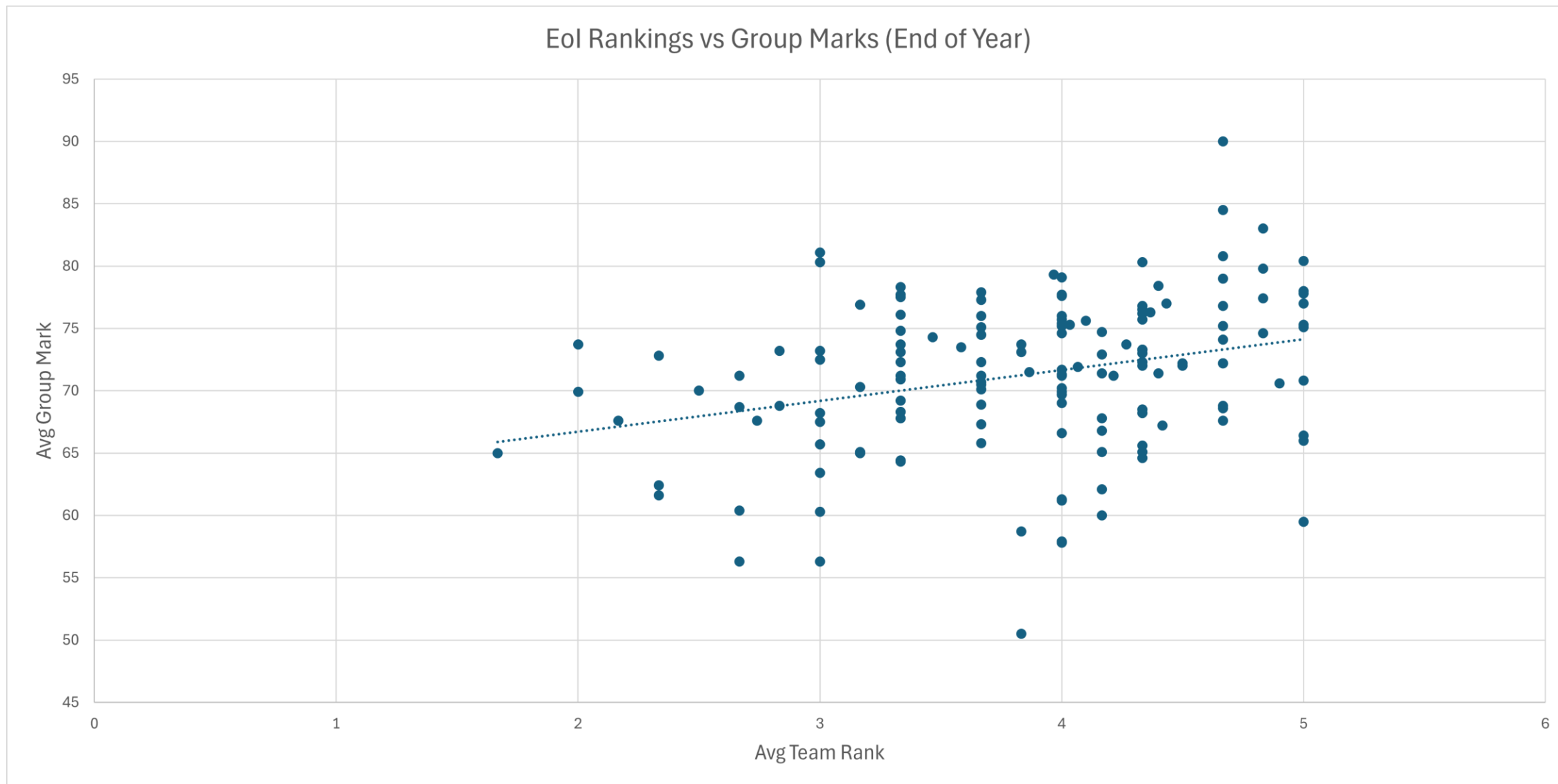
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Team Functioning



Eol Ranking vs. Group Marks (End of Year – since 2021)





- **Have regular meetings with your supervisors / sponsors:**
 - No less than once every 14 days
 - Academic and industry
- **Start writing early**
 - Group / individual reports
 - Peer assessment – at least notes for later
- **Follow a SCRUM process:**
 - Backlog => sprint planning => sprint backlog => stand-ups => sprint review => retrospective => repeat
 - Demo after every sprint



- **Eol feedback for the bids (Eols and pitches) will be made available on Moodle:**
 - Team administrators, please share this with the team
- **The marks on Moodle will be holistic and the average for all three bids**
- **Remember – if there is anything you think I should be aware of, please contact me!**